

Eduardo Dadalto

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Core Expertise

- **AI Safety:** Uncertainty quantification, conformal risk control, confidence calibration, robustness testing, out-of-distribution/anomaly detection.
- **AI Evaluation:** Benchmark design, data collection, large-scale evaluation workflows, data lineage and versioning, performance monitoring, sim2real gap computation, fairness.
- **Experimental Design:** Bayesian hyperparameter optimisation, statistical hypothesis testing, ablation studies.

Professional Experience

Helsing

Paris, France

Sep. 2026 -

Senior AI Research Engineer

- Evaluated multi-agent reinforcement learning (RL) swarming policies and developed an AI assurance methodology for closed-loop simulators.
- Implemented multicalibration methods for vision-language models (VLMs) to improve alignment.

AI Research Engineer

Mar. 2025 - Sep. 2026

- Built AI safeguard models end-to-end, deploying them in the [HX-2](#) product.
- Developed an ONNX library for edge deployment of conformal prediction and confidence calibration methods.
- Implemented scalable workflows for AI product performance and safety evaluation across the company, ensuring robust deployment and fast R&D iteration cycles.

Future Frame

Apr. 2024 - Dec. 2024

Co-founder

San Francisco, CA, USA & Paris, France

- Pre-trained and fine-tuned transformer models on structured data. Released an API platform to customers.
- Raised pre-seed funding from [Transpose VC](#) and [Entrepreneur First](#).

Mila - Quebec AI Institute & International Laboratory on Learning Systems

Oct. 2022 - Jan. 2023

Graduate Research Intern

Montreal, QC, Canada

- Developed misclassification detection algorithms for medical image segmentation.
- Advised by [Pablo Piantanida](#) and [Jose Dolz](#), in collaboration with [Federica](#) and [Balamurali](#).

IRT Saint Exupéry, DEEL Team & Airbus

Apr. 2020 - Oct. 2020

Undergraduate Research Intern

Toulouse, France

- Trained binary-weight neural networks via projected gradient descent for embedded computer vision.
- Supervised by [Franck Mamalet](#) and co-supervised by [François Malgouyres](#) and [Adrien Gauffriau](#) (Airbus).

Education

CentraleSupélec, Université Paris-Saclay, CNRS & IBM Research

Nov. 2020 - Mar. 2024

PhD in Machine Learning, Computer Science Department

Paris, France

- Thesis: [Improving artificial intelligence reliability through out-of-distribution and misclassification detection](#).
- Keywords: Safety in AI, Out-of-Distribution Detection, Uncertainty Estimation, Computer Vision, NLP.
- Advisors: [Pablo Piantanida](#) and [Florence Alberge](#). Committee Members: [Nicolas Vayatis](#), [Yann Chevaleyre](#), [Yves Grandvalet](#), and [Nicolas Thome](#).

Institut de Mathématiques de Toulouse, Université Toulouse III

Sep. 2019 - Oct. 2020

MSc in Applied Mathematics

Toulouse, France

Dissertation: "Training Binarized Deep Neural Networks." Advisors: [Franck Mamalet](#) and [François Malgouyres](#).

ISAE-SUPAERO

Sep. 2018 - Apr. 2020

Diplôme d'Ingénieur, Major in Data & Decision Sciences

Toulouse, France

Selected Coursework: Supervised, Unsupervised, Deep, and Reinforcement Learning with [Emmanuel Rachelson](#). Double Degree with a full tuition fee waiver scholarship.

Instituto Tecnológico de Aeronáutica (ITA)

Feb. 2015 - Jul. 2018

BE in Aerospace Engineering

São José dos Campos, Brazil

Dissertation: "Machine Learning Applied to Communication Channels." Advisor: [Meryem Bennamar](#).

Publications

For a full list of publications, please visit my [Google Scholar page](#).

C: Conference · J: Journal · W: Workshop

- C1. [Eduardo Dadalto](#), [Marco Romanelli](#). *Optimal Zero-shot Regret Minimization for Selective Classification With Out-of-Distribution Detection*. **UAI 2025**. [[link](#)]
- C2. [Eduardo Dadalto](#), [Marco Romanelli](#), [Georg Pichler](#), [Pablo Piantanida](#). *A Data-Driven Measure of Relative Uncertainty for Misclassification Detection*. **ICLR 2024**. [[ARXIV](#)] [[PDF](#)] [[CODE](#)]
- J3. [Eduardo Dadalto](#), [Florence Alberge](#), [Pierre Duhamel](#), [Pablo Piantanida](#). *Combine and Conquer: A Meta-Analysis on Data Shift and Out-of-Distribution Detection*. **TMLR 2024**. [[ARXIV](#)] [[PDF](#)] [[CODE](#)]
- C4. [Maxime Darrin](#), [Guillaume Staerman](#), [Eduardo Dadalto](#), [Jackie CK Cheung](#), [Pablo Piantanida](#), [Pierre Colombo](#). *Unsupervised layer-wise score aggregation for textual ood detection*. **AAAI 2023**. [[ARXIV](#)] [[PDF](#)]
- C5. [Pierre Colombo](#), [Eduardo Dadalto](#), [Guillaume Staerman](#), [Nathan Noiry](#), [Pablo Piantanida](#). *Beyond Mahalanobis Distance for Textual OOD Detection*. **NeurIPS 2022**. [[ARXIV](#)] [[PDF](#)]
- C6. [Eduardo Dadalto](#), [Florence Alberge](#), [Pierre Duhamel](#), [Pablo Piantanida](#). *Igeood: An Information Geometry Approach to Out-of-Distribution Detection*. **ICLR 2022**. [[ARXIV](#)] [[PDF](#)] [[CODE](#)]

Patents

- P1. [Eduardo Dadalto](#), [M. Donini](#), [G. Detommaso](#). *Method and System for Improving Object Tracking*. Filed. 2025.
- P2. [F. Di Salvo](#), [M. Donini](#), [Eduardo Dadalto](#). *Method and System for Conformal-aware Multi-Object-Tracking*. Filed. 2026.
- P3. [Eduardo Dadalto](#), [M. Donini](#). *Method and System for Confidence Multicalibration of Hierarchical Object Classifiers*. In Legal Review. 2026.
- P4. [F. Di Salvo](#), [M. Donini](#), [Eduardo Dadalto](#). *Method and System for Ontology-Guided Collapse of Conformal Prediction Sets*. In Legal Review. 2026.

Preprints

1. [Eduardo Dadalto](#), [Marco Romanelli](#), [Federica Granese](#), [Siddharth Garg](#), [Pablo Piantanida](#). *Trusting the Untrustworthy: A Cautionary Tale on the Pitfalls of Training-based Rejection Option*. 2023.
2. [Eduardo Dadalto](#), [Pierre Colombo](#), [Guillaume Staerman](#), [Nathan Noiry](#), [Pablo Piantanida](#). *A Functional Perspective on Multi-Layer Out-of-Distribution Detection*. 2023.

Workshops

- W1. [Eduardo Dadalto](#), [Marco Romanelli](#), [Georg Pichler](#), [Pablo Piantanida](#). *A Data-Driven Measure of Relative Uncertainty for Misclassification Detection*. **NeurIPS Workshop on Mathematics of Modern Machine Learning**. 2023.
- W2. [Eduardo Dadalto](#), [Florence Alberge](#), [Pierre Duhamel](#), [Pablo Piantanida](#). *Igeood: An Information Geometry Approach to Out-of-Distribution Detection*. **NeurIPS Workshop on Distribution shifts: connecting methods and applications (DistShift)**. 2021.

Teaching

CentraleSupélec, Université Paris-Saclay

Teaching Assistant (TA) for the course *Introduction to Deep Learning (3SQ2050)*.

2021-2023

Gif-sur-Yvette, France

Awards and Scholarships

Nordic Probabilistic AI Summer School ([ProbAI](#))

June 2023

Norwegian University of Science and Technology, Trondheim, Norway

- Coursework on probabilistic modeling, variational inference, and generative models, sponsored by DeepMind.

Globalink MITACS Scholarship

Oct. 2022 - Feb. 2023

Awarded for a winning project in a four-month full mobility scholarship program, Montreal, QC, Canada

- Partnership between Mila, ÉTS, McGill, Inria, CNRS, and Université Paris-Saclay.

BCG Gamma Data Science Hackathon

February 2020

Organized by BCG Gamma and Instituto Tecnológico de Aeronáutica, São José dos Campos, Brazil

- Fine-tuned a YOLOv3 model for object detection on infrared drone footage.
- Received the Best Presentation Award. Advised by [Filipe Verri](#).

Services & Volunteering

- **Reviewer:** Neural Information Processing Systems (NeurIPS 2022, 2023, ...), Computer Vision and Pattern Recognition (CVPR 2023), International Conference on Learning Representations (ICLR 2024, ...), Conference on Artificial Intelligence (AAAI 2023).
- **Presentations:** Experienced at giving talks to large audiences at academic conferences and industry events.
- **Teaching:** Volunteered as a high school math teacher at a Brazilian public school (2011).

Research Grants

- Compute research grant (AD011012803R) at [Jean-Zay](#) a French HPC/AI cluster. 2021 - 2024
- PhD funding from the PSPC AIDA project (2019-PSPC-09) granted by [BPI-France](#). 2020 - 2024

Open Source

[Detectors](#)

May 2023

Open source package to accelerate research on out-of-distribution (OOD) detection for computer vision applications.

[Todd](#) / [ToddBenchmark](#)

February 2023

Open source package to accelerate research on out-of-distribution (OOD) detection for textual applications.

Skills & Tools

- **Languages:** English (fluent), French (fluent), Portuguese (native language), German (notions), Spanish (notions).
- **Programming:** Python (PyTorch, Numpy, Scikit-Learn, Warp, Polars, Optuna, etc.), Go, C, CUDA (basics), JavaScript, HTML, SQL, $\text{\LaTeX}$, git, Linux, agentic AI tools, Large-scale experiment design (Slurm, Metaflow, ArgoCD), Edge deployment (Rust, ONNX, TensorRT), Cloud (Docker, Terraform IaC, AWS, GCP).

Miscellaneous

- **Certificates:** Cambridge C2 Proficiency.
- **Academic contests:** Brazilian Astronomy Olympiad ($3\times$ gold medalist). UN mock debating at FAAP/SP.
- **Side projects:** Co-created [Prompt Riddle](#), an online indie enigma-solving game with 5k+ players.
- **Sports:** brazilian jiu-jitsu, futsal.
- **Rocketry:** Launched and recovered a rocket that reached a 10,000 ft apogee as part of the ITA's Rocket Design team during the 2017 [IREC](#) competition (placed 7th/109 teams) in Las Cruces, NM, USA, which was sponsored by Virgin Galactic. I was responsible for the aerodynamics of the rocket and helped build the GPS localization module.